

◦ Aim:

- To determine the radius of curvature of a planoconvex lens.

◦ Apparatus:

- Sodium lamp, collimator, glass slab, planoconvex lens, mirror, telescope

◦ Theory:

- A planoconvex lens is placed with its convex surface on a mirror so as to enclose a thin film of air between the two. Monochromatic light is focused on a glass slab, placed at an angle of 45° to vertical. A collimator is used to convert light rays from different angles into parallel rays. (check figure 1).

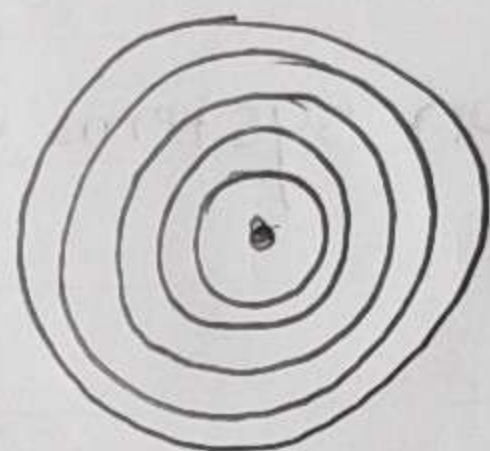
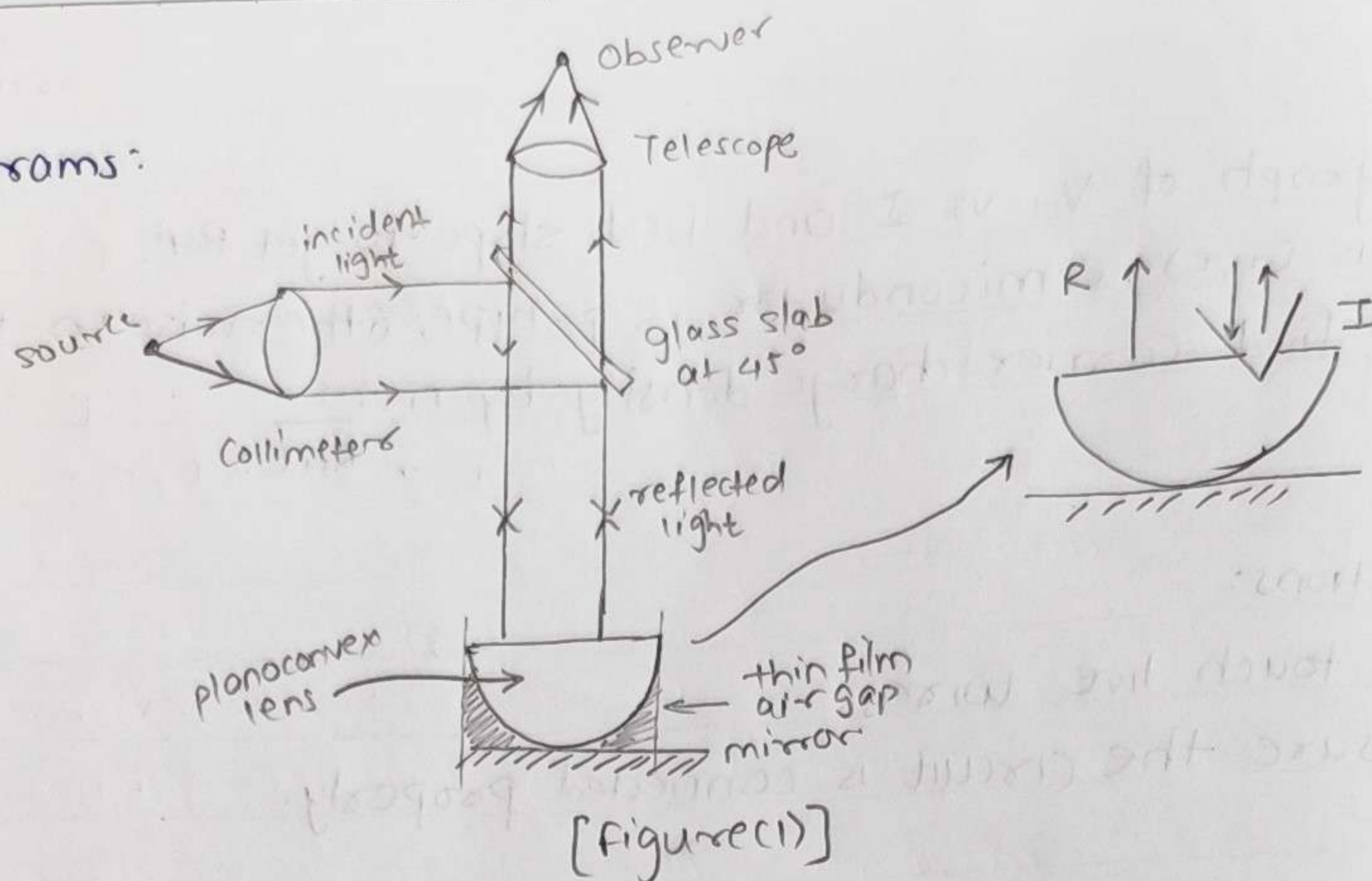
- When the parallel rays hit the glass slab, some rays get transmitted and some get reflected down towards the planoconvex lens. The reflected light first passes through the lens and then the air film and will directly reflect by the mirror. This reflected light goes back to the glass slab, where transmittance & reflectance takes place.

- The transmitted light can be observed through the telescope.

- As we go away from the center, the path difference differs, resulting in phase difference b/w incident & reflected rays. This results in an interference pattern.

- As lens is in contact with the mirror, thickness of air film is zero and hence center is dark. As we move away from center, thickness of air film gradually increases.

Diagrams:



[Figure (2)] - Newton's Rings.

Observation Table:

Ring No.	Left Reading 'a' (cm)	Right Reading 'b' (cm)	Diameter 'D _n ' $D_n = a - b $ (cm)	D_n^2 (cm ²)
2	3.942	3.597	0.345	0.1190
4	3.950	3.576	0.374	0.1398
6	3.970	3.548	0.422	0.1780
8	3.991	3.460	0.531	0.2819
10	4.006	3.419	0.587	0.3445
12	4.022	3.409	0.613	0.3757
14	4.038	3.399	0.639	0.4083
16	4.049	3.357	0.692	0.4788
18	4.061	3.325	0.736	0.5416
20	4.090	3.301	0.789	0.6225

• Calculations:

$$(i) R_1 = \frac{D_{20}^2 - D_2^2}{4(20-2)\lambda} = \frac{0.6225 - 0.1190}{(4)(18)(5893 \times 10^{-8})} = 118.67 \text{ cm}$$

$$R_2 = \frac{D_{18}^2 - D_4^2}{4(18-4)\lambda} = \frac{0.5416 - 0.1398}{(4)(14)(5893 \times 10^{-8})} = 121.75 \text{ cm}$$

$$R_3 = \frac{D_{16}^2 - D_6^2}{4(16-6)\lambda} = \frac{0.4788 - 0.1780}{(4)(10)(5893 \times 10^{-8})} = 127.60 \text{ cm}$$

$$R_4 = \frac{D_{14}^2 - D_8^2}{4(14-8)\lambda} = \frac{0.4083 - 0.2819}{(4)(6)(5893 \times 10^{-8})} = 119.37 \text{ cm}$$

$$R_5 = \frac{D_{12}^2 - D_{10}^2}{4(12-10)\lambda} = \frac{0.3757 - 0.3445}{(4)(2)(5893 \times 10^{-8})} = 126.18 \text{ cm}$$

$$\therefore R_{avg} = \frac{R_1 + R_2 + R_3 + R_4 + R_5}{5} = 122.71 \text{ cm}$$

(ii) By graph, slope = 0.028383.

$$\therefore 0.028383 = 4R\lambda.$$

$$\Rightarrow R_g = \frac{0.028383}{(4)(5893 \times 10^{-8})} = 120.41 \text{ cm}.$$

$$(iii) \text{ Hence, } R = \frac{R_{avg} + R_g}{2} = \frac{122.71 + 120.41}{2} = 121.56 \text{ cm}.$$

- As air film is symmetric about the center, the fringes produced are concentric circles (as shown in Figure c2).

- The diameter of n^{th} concentric circle is given as:

$$D_n^2 = 4nR\lambda \quad \text{--- (i)}$$

Where n is the order of fringe (n^{th} ring).

R is the radius of curvature of planoconvex lens.

λ is the wavelength of source light.

Also, $D_m^2 = 4mR\lambda \quad \text{--- (ii)}$

Subtracting (i) and (ii), $D_n^2 - D_m^2 = 4(n-m)R\lambda$

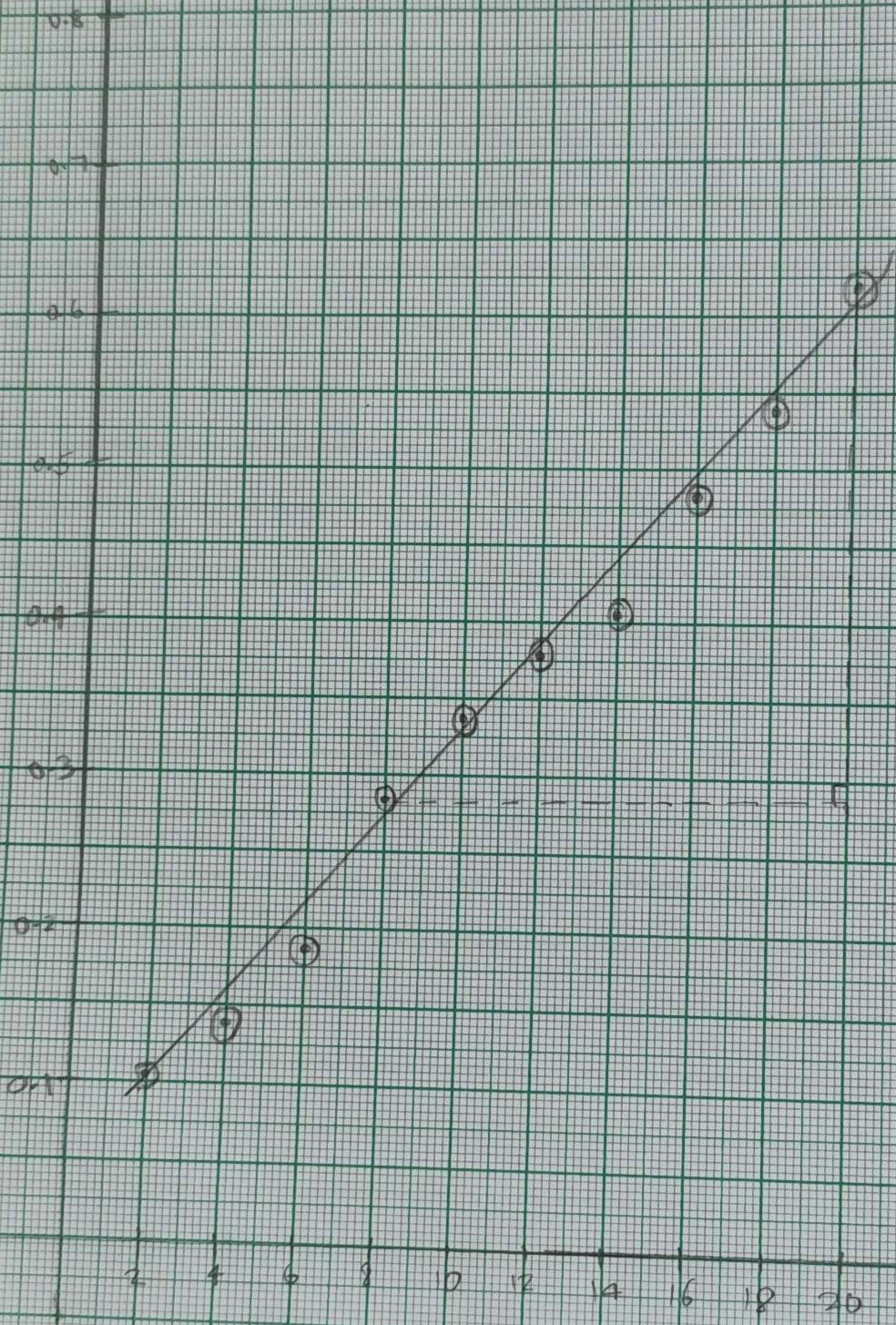
Hence, $R = \frac{D_n^2 - D_m^2}{4(n-m)\lambda}$

• Procedure:

- Arrange the apparatus as shown in figure (c1).
- Switch on the sodium lamp & make sure light falls parallel to slab.
- Adjust the microscope such that it lies exactly above the center of lens and adjust its focal length to observe the rings clearly.
- Using the micrometer screw, move the microscope to the left such that its vertical crosswire is tangential to extreme end of the 20^{th} dark ring and note the readings.
- Move the telescope to the right to take readings of $18^{\text{th}}, 16^{\text{th}}, \dots, 4^{\text{th}}, 2^{\text{nd}}$ rings and repeat the same for right side of centre.
- Find diameter of each ring by subtracting the corresponding rings readings.
- Find radius of curvature using: $R = \frac{D_n^2 - D_m^2}{4(n-m)\lambda} \quad (\lambda = 5893 \text{ \AA})$.

y
 Dn^2
 (cm^2)
 $\uparrow$

Scales on x-axis, 1cm = 2 rings
 on y-axis, 1cm = 0.05 cm²



$$\begin{aligned}
 \text{Slope} &= \frac{0.6225 - 0.2025}{20 - 8} \\
 &= \frac{0.42}{12} \\
 &= 0.035
 \end{aligned}$$

- Take average R of all readings.
- Plot the graph of D_n^2 vs n and find the slope.
 $D_n^2 = 4n\lambda R \Rightarrow \text{slope} = 4\lambda R$. Find R using slope.

Precautions:

- The experiment must be performed in a dark room.
- Make sure to keep a count of the rings.

Applications:

- Newton's Rings is used to control the thickness of paint that is used on posters.

Results:

- Radius of curvature by calculations, $R = 122.71 \text{ cm}$
- Radius of curvature by graph, $R = 120.411 \text{ cm}$.

Conclusions:

- Newton's Rings are observed.
- Radius of curvature of given planoconvex lens is found out to be 121.560 cm.